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Mini Project Report – BAI506

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Mini Project Title
Explainable Tuberculosis Screening from Chest X-Rays with Lesion Localisation for Triage in Resource-Limited Settings

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Project Category	Computer Vision – medical imaging (P2 component report: classifier, evaluation and calibration)

Signature of Guide

Abstract

Tuberculosis remains one of the deadliest infectious diseases, and India carries a quarter of the global burden. Chest radiography is the recommended first screening test, and since 2021 the WHO has endorsed computer-aided detection (CAD) [13] as an alternative to human reading for people aged 15 and above. Published CAD methods report above 0.99 AUROC on the public TBX11K benchmark. This project asked what survives when such a model is evaluated the way a screening programme would use it: on a programmatic active-case-finding cohort with a microbiological reference standard and a certified commercial reader scored on the same patients.

We built a classifier from a frozen radiology foundation model (RAD-DINO) and a 769-parameter regional head that takes the maximum over a 12×12 grid of region scores, so that the region which decides each image is known exactly. It reaches 0.9368 AUROC on TBX11K development and 0.7242 on the reserved half of a 2,708-participant Vietnamese screening cohort, against 0.7862 for Lunit INSIGHT CXR on the same patients. The loss concentrates in people with previous tuberculosis, reproducing a published effect. We then ran eleven pre-registered interventions. Evaluated against their declared decision criteria, several were negligible or adverse and one was positive but below the threshold on the primary split. Three—training to the active-TB label, a related encoder, and fusing a twelve-field screening questionnaire—improve the screening result and leave the benchmark unchanged, and they overlap rather than add. The best configuration reaches 0.7941 with the questionnaire, numerically above the comparator (0.7780) with a confidence interval that includes zero; a field ablation shows the margin above the comparator is a per-scanner adjustment, which unlabelled centering recovered (0.7795 with 25 images per scanner; paired +0.0281 [+0.0074, +0.0486] at 100) under the nominal sampling condition tested, and which fails a pre-registered stability bound when the calibration sample comes from a different population. The decision region of the head overlaps an annotated active-TB lesion in 95.9% of 49 development images. A concept-probing instrument built on radiology-report text embeddings was shown to be negation-blind and its findings withdrawn.

On an exposed screening cohort the system reached a numerically higher AUROC point estimate than commercial CAD; the paired interval includes zero, so no superiority, equivalence or non-inferiority claim is supported. It has a measured low-memory inference footprint on a consumer laptop GPU, exact attribution and a measured localisation result, and the report states why no superiority claim is available from public data.

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1. Introduction

1.1 Background

India accounts for roughly 25% of the world's tuberculosis cases (WHO Global Tuberculosis Report 2025 [12]). Radiologists are scarce relative to need and concentrated in cities, so chest X-rays from district hospitals and screening camps can wait days for a reading. CAD software can read a digital chest X-ray in seconds and is WHO-recommended as an alternative to human reading for screening and triage in people aged 15 and above, against a target product profile of 90% sensitivity at 70% specificity [13].

1.2 Problem statement

As proposed: *to develop an explainable computer-aided detection system that identifies tuberculosis on chest X-rays, distinguishes it from other lung abnormalities, localises the affected regions, and operates at a sensitivity and specificity suitable for triage in resource-limited screening settings.*

As implemented by this component: to develop a chest-X-ray-based tuberculosis screening system that estimates active-TB suspicion, provides region-level attribution, and supports referral decisions in resource-limited screening settings. The production classifier is trained on healthy versus tuberculosis images and is a screening classifier, not a differential-diagnosis model; distinguishing TB from other lung abnormalities was measured (Section 4) but is not what the classifier was built to do.

1.3 Objectives as proposed, and how each stands

proposed objective	outcome
Classifier on TBX11K reaching ≥ 90% sensitivity at ≥ 70% specificity on a held-out internal split, with AUROC reported	The numerical target was achieved on the development subset (sensitivity 1.00 at 70% specificity; AUROC 0.9368 regional head / 0.9949 global probe), **but this is not an independent held-out evaluation**: development data informed early stopping and threshold fitting, the split is provisional, and the internal-test partition was used only once in a rehearsal. **Not met on the screening cohort**: sensitivity 0.62 raw, 0.74 with questionnaire fusion, at 70% specificity.
Three-class classifier (healthy / sick-non-TB / TB)	Measured against a binary head: no difference on the primary metric (0.9661 both). Binary adopted; the three-class output is auxiliary.
Evaluation on Shenzhen and Montgomery	Done: 0.9690 / 0.9690 (regional head). These are curated case-control sets and, as Chapter 4 shows, overstate screening performance by 0.2 AUROC.
TB-region detector and detection AP; TBX11K online challenge submission	Detector is P1/P3's component (not reported here). No challenge submission was made.
Web tool showing TB probability, lesion localisation and ranking; < 2 s per image on a laptop GPU	Served: `/v2/batch` ranks a batch by a TB **screening score** (the image logit—not a probability: a sigmoid of it has expected calibration error 0.49 and calibration slope 0.30 on the Vietnam cohort, so the UI never shows one), with the 12×12 region map, the exact decision region and an overlay per case; triage screen at `/v2/ui`. **Met after the segmenter was moved to the GPU: median 0.13 s/image end-to-end (p90 0.15 s) on the laptop GPU, after a ~40 s first-call model load; masks agree with the training-time CPU masks at 99.998% of pixels.** On CPU alone the pipeline is ≈ 2.1 s/image.

1.4 Scope of this report

This is the P2 component report: it covers the classification and evaluation component in full and references the other components where they interact. If BAI506 requires a single complete team report, the P1 (data and splits), P3 (lesion detector) and P4 (sealed-evaluation machinery) components must be added before submission.

2. Literature Survey

Benchmarks and architectures. TBX11K [1] is the largest public TB chest-X-ray benchmark (11,200 images with lesion boxes). Its authors' SymFormer reaches 99.1% AUROC on the official test split. Shenzhen and Montgomery [2] are the standard external sets. Published numbers on these are near saturation: in our own measurements ten preprocessing variants of one model all reach sensitivity 1.0 at 70% specificity on TBX11K, so the benchmark cannot rank methods finely. General-purpose vision–language models reach about 0.80 zero-shot on TBX11K [11].

Foundation models for chest X-rays. RAD-DINO [8] is a self-supervised ViT-B/14 trained on chest radiographs without text supervision; MAIRA-2 released a continuation trained on a larger corpus. XraySigLIP [10] is an image–text model at 512 px. Frozen encoders with light heads are established for TB: RADAR [4] cross-attends a learned TB query over patches with adversarial domain objectives; CheXFound/GLoRI [5] studies global-plus-local adaptation. We claim no architectural novelty over these.

Real-world evaluation. Codlin et al. [3] evaluated Lunit INSIGHT CXR versions on 2,708 participants of a Vietnamese active-case-finding programme with Xpert Ultra reference, and released the data; they report, as did Kagujje et al. [6] and Qin et al. [7], that CAD accuracy falls in people with previous tuberculosis. Our work reproduces that effect rather than discovering it.

Explainability. Saliency and post-hoc heatmaps are the norm. Max-pooled multiple-instance heads make attribution exact by construction, which is the route taken here. Text-anchored concept probing through vision–language models (CLEAR [9]) was attempted and is reported as a negative result in Chapter 6.

2.1 Limitations of the surveyed work

Benchmark results are reported on curated case-control data with saturated metrics; external validation is usually on two small curated sets; previous-TB performance is rarely stratified; and commercial systems are evaluated mostly by their makers. None of the surveyed academic systems reports a head-to-head against a commercial reader on the same screening participants.

2.2 Outcome of the survey

Evaluate along a ladder from benchmark to screening cohort with a commercial comparator on identical patients; pre-register every intervention; make attribution exact rather than post-hoc; state what the evidence does not support.

3. Data and Methodology

3.1 Datasets

dataset	*n*	role
TBX11K (core train/val lists)	8,400 (800 TB; boxes labelled active or obsolete)	training and internal evaluation; re-split 5,941 / 659 / 899 / 901 (provisional C01)
Vietnam screening cohort (Dryad, [3])	2,708 (280 Xpert-positive; 36% previous TB; three scanners; 12-field questionnaire; Lunit scores)	the programmatic screening cohort used for exploratory evaluation; split once into TUNE 1,352 and REPORT 1,356
Shenzhen, Montgomery	662, 138	curated external
TBX11K DA / DB	72, 104	curated external

Training rows for the production head are the healthy and tuberculosis images of the train split (3,241; 541 positive). Sick-non-TB train images are excluded (Section 4 discusses the consequence).

3.2 Pipeline

1. **Standardise**: greyscale, resize to 512 px (DICOM: VOI LUT, 0.5/99.5 percentile clip).
2. **Quality gate**: a rule-based check that rejects undecodable or implausible inputs; a rejected image never carries a score.
3. **Lung mask**: a public PSPNet segmenter on the GPU; pixels outside the lungs are set to the image mean. This removes a border-reading shortcut quantified in Chapter 4.
4. **Encode**: frozen RAD-DINO ViT-B/14 at 518 px; the 37×37 patch tokens are average-pooled to a 12×12 grid of 768-d region vectors, each L2-normalised.
5. **Head**: one Linear($768 \to 1$) applied to every region; the image logit is the **maximum** over the 144 region logits. Five seeds are averaged.
6. **Operating points**: thresholds at 0.70 and 0.90 specificity, fitted on TBX11K development only.

7. **Optional fusion**: logistic regression on the standardised image score plus twelve questionnaire and scanner fields.

Training: AdamW 3e−3, batch 256, weight decay 10^{-4}, logit adjustment by the training prior, early stopping on development AUROC with patience 15, seeds 17/29/43/61/79. Trainable parameters: 769 per seed (3,845 total) plus 13 for fusion; frozen: 86,580,480.

3.2.1 Baseline versus research candidate v3

component	baseline	research candidate v3
encoder (frozen)	`microsoft/rad-dino` @ `110cbc18...`	`microsoft/rad-dino-maira-2` @ `a6132efb...` (same ViT-B/14 architecture, continued self-supervised training)
training target	`tb_label`: active, latent and mixed TB positive; healthy negative	`active_tb_label`: active and mixed TB positive; latent (obsolete-only) TB **relabelled negative** alongside healthy
training rows	3,241 (541 positive)	3,241 (447 positive)
region aggregation	12×12 maximum	12×12 maximum
head	Linear($768 \rightarrow 1$), 769 parameters per seed, 5 seeds	identical
clinical fusion	optional, 12 fields	optional, 12 fields including two scanner indicators
status	served default (bundle v2)	frozen research candidate (bundle v3), not promoted

3.3 Evaluation protocol

Every experiment after the headline result was pre-registered: a protocol file hashed before any fitting, a primary evaluation split that the intervention was not designed on (TBX11K calibration, 899 images, 81 active-TB positives), a minimum meaningful effect of +0.02 AUROC with a paired-bootstrap interval excluding zero, and a reproduction gate requiring the production arm to be reproduced to within 0.0005 before any contrast is read. The screening tuning half is reported as context and is exposed (Chapter 7). Interventions that act on cohort-specific quantities (scanner subspace, aggregator selection, local-site refit, questionnaire fusion, per-scanner centering) have no calibration endpoint by design and are exploratory on the screening cohort. Bootstrap resampling is at the participant level; the released data carry no screening-event identifier,

so clustering within the 127 screening events is not modelled and intervals may understate uncertainty. TBX11K's internal-test partition and the Vietnam REPORT half were not read for any of the interventions.

3.4 Software architecture

```
image (any size) -> standardise 512x512 -> quality gate -> reject | pass
  -> lung mask (PSPNet, GPU; outside-lung -> image mean)
  -> frozen encoder (ViT-B/14 @ 518 px) -> 37x37 tokens -> pool -> 12x12 x
768-d, L2
  -> Linear(768->1) per region, 5 seeds -> max over 144 regions -> image
logit
  -> operating points (dev-fitted) -> optional fusion (12 fields,
exploratory)
served by scripts/serve_v2.py:  /v2/analyse, /v2/batch (ranked),
/v2/overlay, /v2/ui, /v2/calibration/propose
```

Figure 3.1: The served pipeline. The arg-max region is the decision region exactly.

4. Results I: the Ladder from Benchmark to Clinic

4.1 The benchmark figure is partly a shortcut

A global linear probe on the frozen encoder reaches 0.9949 on TBX11K development—and 0.9094 from the **outer 12% of the image alone**, a frame containing no lung. The shortcut is dataset-specific: the border-only model scores 0.52–0.62 at the four external sites, while a lungs-only model tracks the full image within 0.015 everywhere. The production model is therefore lung-masked.

4.2 Performance along the ladder

rung	cohort	regional 12×12 max	comparator
benchmark	TBX11K development (659)	0.9368	—
curated external	Shenzhen / Montgomery	0.9690 / 0.9690	—
curated external	DA / DB	0.9755 / 0.9497	—
screening	**Vietnam REPORT (1,356)**	**0.7242** [0.6843, 0.7618]	**0.7862**

The aggregator matters: `mean` over the same regions (mathematically global pooling) stays at 0.636–0.642 on the tuning half while `max` reaches 0.7239; learned gated attention is worse (0.6416). Selection optimism between tuning and reporting halves is −0.0005.

The screening loss is a negative-class problem. TB-versus-healthy on TBX11K is AUROC 1.0000; a screening queue's Xpert-negatives include healed TB and chronic lung disease, and the model learned abnormality. A distance-based out-of-distribution gate does not see this: the screening cohort sits at the same embedding distance as Shenzhen and the gate passes 99.2% of it.

4.3 Where the loss is: previous tuberculosis

Under the construction used in the published literature, our AUROC is 0.7720 in participants without previous TB and **0.6392** with it (degradation +0.1328 [0.0718, 0.1951]). The comparator degrades from 0.8170 to 0.7192. The difference in differences, +0.0481 [−0.0170, +0.1113], includes zero: we cannot claim to degrade more than the comparator. Scanner appearance is not the cause: removing the scanner-predictive subspace from the embedding changes AUROC by −0.0002.

5. Results II: Interventions

#	intervention	calibration Δ (primary)	screening Δ (context)	verdict
1	scanner-subspace removal	—	−0.0002	null
2	gated-attention aggregator	—	−0.082	worse
3	**active-TB label** (latent → negative)	−0.0037	+0.0203 **[**+0.0127**,** +0.0279**]**	external only
4	head refit on local labels vs threshold only	0 of 30 helped	—	null
5	lesion-box region supervision	+0.0019 [−0.0019, +0.0061]	−0.0108	null
6	**clinical-context fusion**	—	+0.058 **[**+0.030**,** +0.085**]**	external only
7	three training sources	+0.0033 [+0.0006, +0.0064]	+0.0036	sub-threshold
8	encoder ensemble	+0.0006	+0.011	null
9	**MAIRA-2 encoder**	−0.0004	+0.0188 **[**+0.0015**,** +0.0364**]**	external only
10	levers 3 + 7 + 9 stacked	−0.0010	+0.0286 (predicted +0.047)	not additive
11	label-free scanner centering	—	+0.0281 [+0.0074, +0.0486] (n = 100)	worked under nominal sampling; fails stability bound

The three external levers overlap. The best image-only head on the screening cohort is MAIRA-2 with the active label: 0.7585 [0.722, 0.796], +0.0348 over production, −0.0032 on calibration. Adding the extra sources on top subtracts. A 2,304-dimensional head on three concatenated encoders has the best development AUROC ever fitted (0.9505) and the worst transfer (−0.0137): more capacity learned the benchmark better and the clinic worse, which is why the backbone was never fine-tuned.

Clinical-context fusion (10-fold × 20 repeats, tuning half):

image score	raw	fused	previous TB raw → fused
production head	0.7237	0.7809	0.6144 → 0.7122
MAIRA-2 + active head	0.7585	**0.7941**	0.6501 → 0.7349
Lunit	0.7780	0.7778	0.7062 → 0.7144

Fused MAIRA-2 against the comparator's raw score: +0.0170 [−0.0193, +0.0527]—numerically above, not statistically. The comparator keeps the better AUPRC (0.328 vs 0.300). Sensitivity at 70% specificity: 0.741 vs 0.708.

What carries the fusion gain. Leave-one-field-out: only the **scanner identifier** costs with an interval excluding zero (−0.0138; −0.0429 in previous TB). With the ten questionnaire fields alone the fused model is 0.7803 against the comparator's 0.7780, within noise. In particular, the previous-TB improvement in the fusion table above is carried mostly by the scanner term, not by the `past_tb` field (removing it changes the previous-TB AUROC by +0.004): the questionnaire does not independently solve the previous-TB deficit. Scanner indicators produced a substantial improvement in our candidate (≈ 0.02 overall, 0.04 in previous TB), while adding the same fields to the comparator produced no material improvement in this experiment (+0.001). This does not independently establish scanner invariance in either system.

Label-free scanner calibration. Centering each scanner's scores on an unlabelled batch of its own images recovers the supervised gain—0.7795 with 25 images per scanner, and at 100 images a paired +0.0281 [+0.0074, +0.0486] (previous TB +0.0557) over the raw score—under the nominal sampling condition tested, and equalises referral burden across scanners (31% / 24% / 50% → 40% / 31% / 32%). It fails a pre-registered stability bound when the calibration batch is drawn from a different population: a previous-TB-only batch lowers that scanner's sensitivity by 0.122 (bound 0.10). The supervised term fails worse under the same shifts. The method is reported as not usable without a consecutive-sample guard, whose safety would have to be shown prospectively; a fail-closed proposal endpoint implements the guard and is labelled experimental.

Cost. Encoder 0.030 s/image on a 6 GB laptop GPU (peak 0.69 GB), 1.25 s on CPU; head fit 6.6 s; full pipeline with the segmenter on the GPU: median 0.13 s/image, p90 0.15 s (masks identical to the CPU-built training cache at 99.998% of pixels); ≈ 2.1 s/image on CPU. Threshold recalibration to a new site needs about 25 labelled local images.

6. Explainability

6.1 Exact attribution, measured

Because the image logit is the maximum over region logits, the arg-max region is the region that decided—no gradient, no post-hoc map. The attribution is exact at the 12×12 region level (cells of about 43 px on a 512 px image), not at pixel-level lesion boundaries. The unit of evaluation is the pooled 12×12 region cell: a cell overlaps a lesion box if its pixel footprint shares at least one 14-px ViT patch with the box after the exact preprocessing chain. On the 49 development images carrying an active-lesion box, the decision cell overlaps a box—a top-1 region-cell hit—in **95.9%** [89.8, 100] (top-3 cells: 100%), against 36.7% for a random lung cell, 18.4% for a random cell and 8.2% for the centre cell; the box-supervised head's top-1 region-cell hit rate on the same images is 0.939. This is "the strongest region is on a lesion", not lesion segmentation; it is exploratory on 49 curated images.

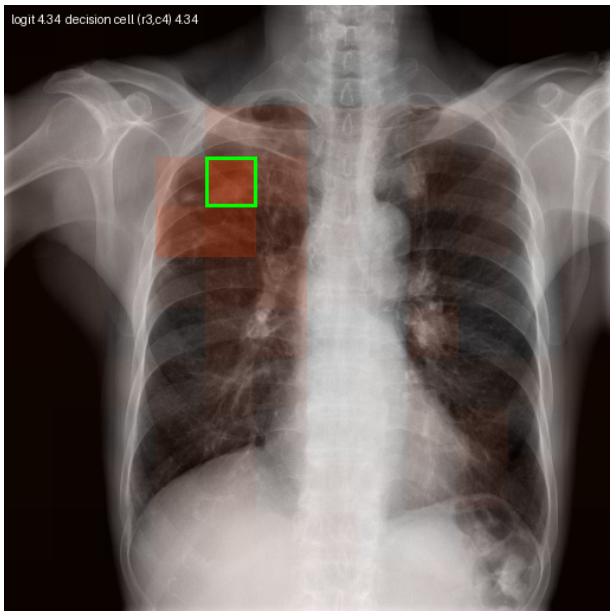


Figure 6.1: Served overlay for a development active-TB image (logit 4.34): region logits shaded, the decision cell (row 3, column 4) boxed. Region-cell resolution, not a lesion boundary.

6.2 A negative result on concept probing

To ask what the score responds to, we correlated it with similarity to single-phrase text anchors through a public radiology vision–language model. *Scarring* correlated at $\rho = 0.566$ with our score while carrying no information about TB (AUROC vs Xpert 0.494); *consolidation* correlated negatively. A forensic pass showed the instrument cannot separate a finding from its negation:

no scarring correlates with the score at +0.301, the same sign as *scarring*; *no consolidation* at −0.534, more negative than *consolidation*, because 27.6% of vocabulary sentences about consolidation are negated normal-report language. All mechanistic readings from this instrument are withdrawn; the method limitation is reported instead.

7. Limitations and What the Evidence Supports

- **Every screening number is on an exposed cohort.** The tuning half was used for selection by design; the reporting half, intended as read-once, was scored in six analyses, four of which influenced a decision. The three external levers were each found on the tuning half, so the fused 0.7941 is a best case designed on the cohort.
- **The TBX11K split is provisional** and ships no patient identifier; 23 of 899 calibration images have a cross-split near-duplicate (effect 0.0003).
- **No superiority, equivalence or non-inferiority is established** against the commercial comparator; the paired interval includes zero and the margin above the comparator is a scanner intercept.
- **Confidence intervals are participant-level**; no screening-event identifier is released, so within-event clustering is not modelled.
- **Scanner invariance is not demonstrated**; the label-free calibration is investigated, not validated.
- **The localisation result** is on 49 development images; the lesion detector proper is another component.
- HIV status was unavailable in every cohort.

What the evidence supports: *on an exposed screening cohort, a frozen-encoder regional head with questionnaire fusion achieved a numerically higher AUROC point estimate than commercial CAD, with a paired interval that includes zero; without the scanner term the point estimate is 0.7803 against 0.7780; it runs on a consumer laptop GPU with a low memory footprint; its attribution is exact at region level and lands on annotated lesions in an exploratory test.*

8. Conclusion and Future Scope

A frozen radiology foundation model with a 769-parameter regional head reaches 0.94 on the benchmark and 0.72 on a screening cohort where commercial CAD reaches 0.79. Eleven pre-registered interventions found three levers that move the screening number and none that moves the benchmark; with a twelve-field screening questionnaire the system reaches a point estimate near the comparator's, and with a scanner-associated adjustment a point estimate above it, with paired intervals that include zero in both cases. The honest description is a scanner-aware screening research system with exact attribution, a measured localisation result, and an experimentally investigated label-free calibration mechanism—not a self-calibrating system and not a better one.

Future scope. (1) A head-to-head on a cohort nobody on the team has touched, with the frozen candidate and the comparator scored on the same patients and previous TB as a pre-specified subgroup; the protocol is pre-registered and hashed and a one-shot scorer exists. (2) A prospective test of the consecutive-sample guard for label-free calibration. (3) Documenting the benchmark's cross-split duplicates for the community.

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A. Reproduction

Every number traces to a run directory under `artifacts/` containing a hashed protocol, result tables, a test suite and a SHA-256 manifest; the project record indexes runs R1–R25:

`artifacts/P2_PROJECT_RECORD_20261101.md`

The served system is `scripts/serve_v2.py` (default bundle v2, RAD-DINO; `-bundle artifacts/p2/bundle_v3_candidate` selects the frozen research candidate); the triage screen is at `/v2/ui`. Unit tests: `py -3 -m pytest tests/ -q`.